1.Aim of the practical:

- Create Pivot Tables to summarize large datasets efficiently
- Group data within Pivot Tables (by dates, numbers, or custom categories)
- Create and manage multiple Pivot Tables on a single worksheet

2.Objective:

- To learn how to create, group, and manage Pivot Tables in Excel for efficiently summarizing and analyzing large datasets. This includes generating multiple Pivot Tables on a single worksheet and applying grouping techniques (by date, number, or category) to extract deeper insights.
- To understand the core functionality and structure of Pivot Tables for dynamic data summarization.

3.Code for experiment/practical:

DATASET :

Region	Product	Units Sold	Unit Price	Date	Total Sales
East	Phone	32	602	########	19264
West	Tablet	39	545	########	21255
North	Tablet	49	710	########	34790
East	Monitor	32	346	########	11072
East	Tablet	4	347	########	1388
West	Tablet	30	910	########	27300
North	Tablet	37	688	########	25456
North	Monitor	23	839	########	19297
East	Tablet	39	750	########	29250
South	Phone	45	537	########	24165
East	Monitor	15	840	########	12600
East	Phone	43	978	########	42054
East	Monitor	29	672	########	19488
East	Tablet	36	350	########	12600
West	Phone	13	614	########	7982
North	Monitor	32	497	########	15904
West	Laptop	7	810	########	5670
West	Tablet	22	462	########	10164
West	Monitor	28	963	########	26964
East	Laptop	2	343	########	686
South	Monitor	42	545	########	22890

Create a Pivot Table & Group Data (e.g., by Month)

1. Open the File
- Open the Excel file: random_sales_data.xlsx
2. Select a Cell in the Dataset
- Click anywhere inside the data (e.g., cell A1).
3. Insert a Pivot Table
- Go to the Insert tab → Click PivotTable
- In the dialog:
    - Table/Range: Excel will automatically detect the range.
    - Choose where to place the PivotTable:
4. Set Up the Pivot Table
- The PivotTable Fields pane appears.

Let's say we want to analyze Total Sales by Product and Region:
- Drag Product → Rows
- Drag Region → Columns
- Drag Total Sales → Values

5. Group by Date (Optional - Monthly Grouping)

If you'd rather analyze over time:
- Drag Date → Rows
- Right-click any Date in the PivotTable → Click Group

Sum of Total Sales	Column Labels				
Row Labels	East	North	South	West	Grand Total
Laptop	90228	71136	106627	39215	307206
Monitor	48392	77822	107999	71233	305446
Phone	142109	37566	56987	219299	455961
Tablet	72197	112662	127355	103064	415278
Grand Total	**352926**	**299186**	**398968**	**432811**	**1483891**

Sum of Total Sales	Column Labels				
Row Labels	East	North	South	West	Grand Total
Laptop	**90228**	**71136**	**106627**	**39215**	**307206**
Jan	686	37035	13040	19754	70515
Feb	54282	34101	38538	19461	146382
Mar	35260		22415		57675
Apr			32634		32634
Monitor	**48392**	**77822**	**107999**	**71233**	**305446**
Jan	43160	41025	42822	37993	165000
Feb	5232		25887	30319	61438
Mar		36797	39290	788	76875
Apr				2133	2133
Phone	**142109**	**37566**	**56987**	**219299**	**455961**
Jan	61318		24165	13610	99093
Feb	42620	759	21911	51995	117285
Mar	38171	23451		110622	172244
Apr		13356	10911	43072	67339
Tablet	**72197**	**112662**	**127355**	**103064**	**415278**
Jan	43238	60246	42070	58719	204273
Feb		35841		44345	80186
Mar	28959	16575	71837		117371
Apr			13448		13448
Grand Total	**352926**	**299186**	**398968**	**432811**	**1483891**

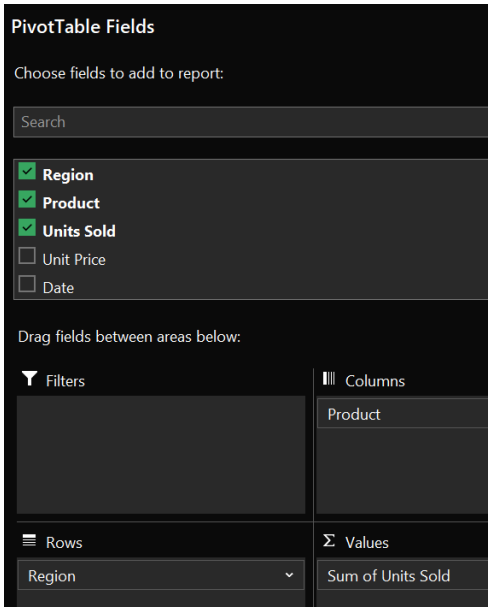
Create Multiple Pivot Tables in One Worksheet

1. Click on the Dataset Again
- Go back to the original data sheet.
- Select any cell inside the data.
2. Insert Another Pivot Table
- Insert → PivotTable
- In the dialog:
    - Choose Existing Worksheet

3. Set Up This Second Pivot Table

- Drag Region → Rows
- Drag Units Sold → Values

Sum of Units Sold	Column Labels				
Row Labels	Laptop	Monitor	Phone	Tablet	Grand Total
East	200	82	200	128	610
North	138	154	74	142	508
South	165	179	93	221	658
West	87	120	417	140	764
Grand Total	**590**	**535**	**784**	**631**	**2540**



4.Result/Output:

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Grand Total	**352926**	**299186**	**398968**	**432811**	**1483891**

Learning outcomes (What I have learnt):

- Create Pivot Tables to quickly summarize large and complex datasets without using formulas.
- Group data within Pivot Tables by.
- Build and manage multiple Pivot Tables on a single worksheet to compare insights side by side.